

| Method      | B@1          | B@2          | B@3          | CIDEr         | METEOR       |
|-------------|--------------|--------------|--------------|---------------|--------------|
| ClipCap     | 29.75        | 16.32        | 9.48         | 125.45        | 19.25        |
| GIT         | 35.19        | 20.12        | 11.90        | 155.38        | 23.06        |
| <b>Ours</b> | <b>37.27</b> | <b>21.92</b> | <b>13.90</b> | <b>174.47</b> | <b>25.07</b> |

Table 2: Baseline comparison on Moments-OVRE.

| CLIP | GPT2 | CIDEr         | METEOR       |
|------|------|---------------|--------------|
| ✗    | ✗    | 131.85        | 19.84        |
| ✗    | ✓    | 165.67        | 24.12        |
| ✓    | ✓    | <b>174.47</b> | <b>25.07</b> |

Table 3: Study on vision and language model fine-tuning. The ✓ mark means fine-tuning the corresponding module.

set to  $1e-6$  for CLIP,  $2e-5$  for GPT-2, and  $1e-3$  for Attention-Pooler. We applied learning rate warm-up during the early 5% training steps followed by cosine decay. We trained the networks for 50 epochs on 8 Nvidia V100 GPUs and chose the model with the highest CIDEr score as the final model.

## Main Results

**Baselines models.** Previous VidVRD methods focused on predicting relationships over detected objects, which is essentially a classification task and thus cannot be applied to OVRE. Therefore, we introduce several generative models as baseline models.

- ClipCap (?) is an image captioning model that utilizes the same visual encoder and text decoder as we do. It uses a mapping network to convert CLIP embeddings into GPT-2 prefixes. To apply ClipCap for videos, we follow the most common strategy that treats each video frame as an individual image and then perform a mean pooling layer along the temporal dimension to obtain a global video representation.

- GIT (?) stands as a vision-language generative model, demonstrating strong performance across numerous generation tasks. This achievement is attributed to its effective optimization of the language model loss during pre-training, involving a substantial collection of image-text pairs. We directly fine-tune  $GIT_B$  to generate relation triplets without making further modifications.

**Result and Analysis.** We present our results on Moments-OVRE in Table 2 and compare our approach with baseline methods trained under the same training settings. Our approach outperforms baseline generative methods, achieving a higher METEOR score (+6.22) than ClipCap and (+2.01) than GIT. We find that although GIT was pre-trained on 0.8B image-text pairs and achieved impressive performance on video captioning datasets, it did not perform as well as our approach on the OVRE task. This could be attributed to the fact that the image-text generative pre-training does not directly facilitate the understanding of fine-grained information such as relationships in videos.

| Features | CIDEr         | METEOR       |
|----------|---------------|--------------|
| Region   | 77.38         | 13.14        |
| Frame    | 115.85        | 18.11        |
| Patch    | <b>165.67</b> | <b>24.12</b> |

Table 4: Comparisons with different visual features (w/o fine-tuning visual encoder).

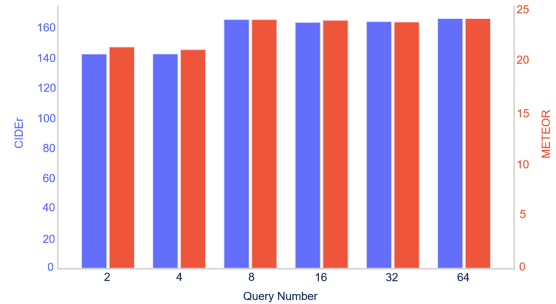


Figure 6: Impact of the query numbers on the performance. For each query number, we report CIDEr (blue) and METEOR scores (red) over the Moments-OVRE test set.

## Ablation Study

**The Impact of Query Numbers.** We first investigate the impact of query numbers. Specifically, we experiment with query number=2, 4, 8, 16, 32, and 64. Figure 6 shows that an extremely small number of queries will yield inferior generation results since the limited queries are insufficient to extract the content in video patches. As the number grows from 2 to 8, the generated results show significant improvement. Subsequently, when the number of queries continues to double, the performance of the model gradually becomes saturated. We choose query number=64 as it demonstrated the best performance across all metrics.

**Exploration of Vision and Language Model Fine-tuning.** Recent research (?) has shown that fully fine-tuned CLIP can effectively bridge the modality gap in the video domain. As shown in Table 3, though the trainable parameters increased, fine-tuning the CLIP-ViT can improve the CIDEr score by 8.8. Additionally, we delve into the outcomes of fine-tuning the text decoder. Experiments reveal that keeping GPT-2’s parameters fixed results in a notable reduction in the CIDEr score. We attribute this decline to that GPT-2 is primarily proficient in generating natural language rather than triplets. Therefore, fine-tuning becomes imperative to enhance its capacity for producing rarely seen triplet sequences.

**The Effect of Different Visual Features.** Our experimental investigations involve ablations on different granularity of visual features. Within our proposed framework, we employ patch features extracted from videos as prefixes for the text decoder. Furthermore, we explore two alternative representations as inputs to the attentional pooler: (I) Region features: Following the common VidVRD practice, we extract a sequence of objects and subsequently employ a tracking algorithm to obtain 5 tracklet features per video. These fea-

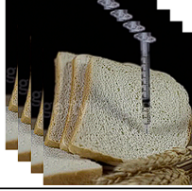
|              |                                                                                                |                                                                                                                                 |                                                                                    |                                                                                     |
|--------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|              |               |                                                |  |  |
| Ground Truth | woman hold knife knife<br>cut meat meat placed<br>on chopping board<br>chopping board on table | hand hold towel water<br>wet towel tower scrub<br>baby baby hold toy duck<br>basin filled with water<br>toy duck float on water | left hand hold handcuffs<br>right hand hold woman<br>handcuffs handcuff woman      | syringe inserted into<br>slice of bread wheat<br>next to slice of bread             |
| ClipCap      | man hold knife knife<br>cut meat meat placed<br>on chopping board<br>chopping board on table   | right hand hold toothbrush<br>toothbrush inside mouth                                                                           | crowd sit on chair<br>crowd look at man                                            | left hand press bread<br>right hand hold knife<br>knife cut bread                   |
| Git          | man hold hammer<br>hammer beat nail<br>nail nailed to wooden board                             | boy hold toy duck<br>toy duck in bathtub                                                                                        | crowd sit on chair<br>crowd look at man                                            | electric drill drill bread                                                          |
| Ours         | man hold knife knife<br>cut meat meat placed<br>on chopping board<br>chopping board on table   | boy sit in bathtub<br>boy hold toy<br>toy immersed in water<br>water in bathtub                                                 | man hold handcuffs<br>handcuffs handcuff woman                                     | syringe pierce bread<br>bread placed on<br>chopping board                           |

Figure 7: Comparisons of triplets generation across diverse OVRE methods. The illustration highlights accurately described triplets in green, triplets with semantic correlation in blue, and irrelevant triplets in red.

tures replace patch features as input to the model. Specifically, we utilize RegionCLIP (?) pre-trained from LVIS to crop bounding boxes and seqNMS (?) for object tracking. (II) Frame features: We directly utilize features extracted from individual frames using CLIP, concatenating them to form a representation of frame-level features. As depicted in Table 4, both frame features and region features exhibit poor performance. Notably, frame features capture the overall visual content of an image but overlook finer details such as objects and relationships. Surprisingly, region features fare even worse compared to frame features. We hypothesize that this is attributed to the limited generalization capability of existing object detectors. The diverse range of object categories complicates their accurate detection within our Moments-OVRE context.

## Conclusion

In this paper, we introduce a new task named OVRE, where the model is required to generate all relationship triplets associated with the video actions. Concurrently, we present the corresponding Moments-OVRE dataset, which encompasses a diverse set of videos along with annotated relationships. We conduct extensive experiments on Moments-OVRE and demonstrated the superiority of our proposed approach over other baseline methods. We hope that our task and dataset will inspire more intricate and generalizable research in the realm of video understanding.

**Limitations:** (I) This version of Moment-OVRE has currently omitted BBox annotation due to the high cost of annotation. We are committed to progressively enhancing this dataset and intend to introduce BBox annotations in upcoming

versions of Moments-OVRE. (II) For extracting action-centric relations, leveraging commonsense among action categories and relations (?) or implicit knowledge-driven representation learning methods (??) have shown promise. We will consider these knowledge-driven methods in future work.

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